Design of Enterprise IoT: Assignment 1a

Smart Turtle Breeding System (STBS)

Team 01

Jinling Institute of Technology

Date: October 21, 2026

Student ID	Name	Preferred Name	Contribution (%)
2422441049	Li Jingru	Leamon	25%
2422441046	Rong Yicheng	Bacon	25%
2422441050	Zhang Zhengyan	Paul	25%
2422441048	Ye Taohao	Leaf	25%

1 Task 1: Use Case Description

A Chinese soft-shell turtle farm operates 20 indoor or semi-closed aquaculture units with an operations and maintenance team. Farm operators manage water quality, feeding, temperature, turtle activity and nesting areas; maintenance technicians service filters and pumps.

Operators manually measure water conditions, prepare feed, inspect behaviour and health, change water, and search for eggs. Periodic rounds can cause inconsistent feeding, delayed detection of deteriorating conditions or equipment failures, and labour-intensive egg collection. Water quality, feeding and temperature affect turtle health and growth [1]. Reviews identify temperature, dissolved oxygen and pH as priority IoT aquaculture parameters [2, 3].

An IoT solution is proposed to monitor behavioural conditions continuously, control feeding and temperature, and support bounded partial water changes. Distributed sensors, actuators and cameras send alerts and records so one team can respond between rounds; field evidence shows that vision-based nesting detection can reduce egg-harvesting effort [4].

2 Task 2: Solution Overview

STBS uses water-temperature, pH, turbidity and ammonia sensors in each unit, with level/flow feedback, feeder sensors and IP cameras covering activity, feeding and nesting zones. Local controllers compare readings with species- and growth-stage set-points, regulate heaters, dispense scheduled portions and perform bounded partial water changes. Device feedback and follow-up measurements verify each action. An edge gateway keeps essential rules operating during internet loss, filters routine data and sends readings, events and selected video to a cloud dashboard. Alerts identify the affected unit, abnormal parameter, action taken and need for intervention. Farm operators configure targets, review alerts

and nesting evidence, collect eggs and override automation. Maintenance technicians calibrate sensors, service filters and equipment, diagnose failures and maintain power/network availability. Role-based access, audit logs and local fail-safe limits support reliable operation.

3 Task 3: Value Proposition

The value proposition follows the tutorial method. It distinguishes the current substitute from the proposed system, separates benefits from features, describes the user experience, and records adoption needs and implementation fears. Each statement identifies the stakeholder to whom it applies.

IoT solution: Smart Turtle Breeding System (STBS) **People: Farm Operators and Maintenance Technicians**	
Substitute: What do people do without employing the solution? • Farm operators have to manually measure the water quality and temperature in each breeding unit. (Farm Operators) • Farm operators have to prepare feed portions and feed turtles according to daily schedules. (Farm Operators) • Farm operators have to regularly inspect turtle activity, feeding behaviour, and health conditions. (Farm Operators) • Farm operators have to inspect nesting areas and search for turtle eggs manually. (Farm Operators) • Farm operators have to change water manually, while maintenance technicians have to schedule filter maintenance based on experience and inspection records. (Farm Operators and Maintenance Technicians)	Benefits: What is good about the solution? • The STBS aims to maintain suitable water conditions and reduce the duration of environmental abnormalities. (Farm Operators) • The system can improve feeding consistency and reduce missed or inaccurate feeding. (Farm Operators) • The system can help detect suspected abnormal activity or feeding behaviour earlier, allowing timely inspection. (Farm Operators) • The system can identify suspected nesting activity and locations, reducing unnecessary searches for eggs. (Farm Operators) • The system can reduce repetitive work and provide operating records for condition-based filter maintenance. (Farm Operators and Maintenance Technicians)
Features: How does the solution work? • The STBS continuously measures water temperature, pH, turbidity, and ammonia nitrogen, and automatically regulates water temperature. (Farm Operators) • The system automatically dispenses scheduled feed portions and verifies delivery through sensor feedback. (Farm Operators) • The system analyses video to identify turtle activity, feeding behaviour, and suspected nesting events, and stores recordings for verification. (Farm Operators) • The system automatically operates pumps and valves for bounded partial water changes and verifies water level, flow, and post-action water quality. (Farm Operators and Maintenance Technicians) • The system connects local controllers to the cloud through a gateway and provides remote monitoring, alerts, operation verification, and manual override. (Farm Operators and Maintenance Technicians)	Experience: What does it feel like to use the solution? • Farm operators will feel more confident because they can monitor all 20 breeding units from a central dashboard. (Farm Operators) • Farm operators will feel less worried because alerts show abnormal readings and the corrective action taken by the system. (Farm Operators) • Farm operators will feel more in control because they can verify automated operations and intervene when necessary. (Farm Operators) • Farm operators will feel less burdened because they can review nesting videos and records before conducting targeted on-site checks. (Farm Operators)
Fears: What are potential risks in employing the solution? (E.g., what may not work, what may go wrong, etc.) • There could be power or network failures that interrupt monitoring, automated control, or remote alerts. (Farm Operators and Maintenance Technicians) • There could be sensor fouling, ageing, or calibration errors that lead to inaccurate readings and control decisions. (Farm Operators and Maintenance Technicians) • The video analysis may falsely identify or miss abnormal behaviour and nesting events. (Farm Operators) • There could be failures of feeders, heaters, pumps, or valves that disrupt feeding and breeding conditions. (Farm Operators and Maintenance Technicians) • There might be high installation and ongoing costs for electricity, calibration, repairs, and consumables. (Farm Operators and Maintenance Technicians)	Needs: What are the hidden needs and rational drivers for employing the solution? • Farm operators need suitable water and temperature conditions for different turtle species and growth stages. (Farm Operators) • Farm operators need to reduce repetitive inspection, feeding, and water-change work so that all 20 units can be managed efficiently. (Farm Operators) • Farm operators need remote monitoring, timely alerts, and confirmation that automated corrective actions were effective. (Farm Operators) • Farm operators need reliable activity, feeding, and nesting records to support on-site inspection and breeding management. (Farm Operators) • Farm operators and maintenance technicians need an affordable, easy-to-maintain system with manual override and complete maintenance records. (Farm Operators and Maintenance Technicians)

4 References

[1] Food and Agriculture Organization of the United Nations. "Cultured Aquatic Species Information Programme: Chinese soft-shell turtle (*Trionyx sinensis*)." Available at FAO cultured species profile.

[2] D. R. Prapti et al. "Internet of Things (IoT)-based aquaculture: An overview of IoT application on water quality monitoring." *Reviews in Aquaculture*, 14, 979–992, 2022. doi:10.1111/raq.12637.

[3] H. Rastegari et al. "Internet of Things in aquaculture: A review of the challenges and potential solutions based on current and future trends." *Smart Agricultural Technology*, 4, 100187, 2023. doi:10.1016/j.atech.2023.100187.

[4] A. Munyaev, B.-L. Jiang, and I.-J. Huang. "Smart turtle farm: Soft-shell turtle nesting behavior recognition and monitoring system." *Computers and Electronics in Agriculture*, 244, 111477, 2026. doi:10.1016/j.compag.2026.111477.